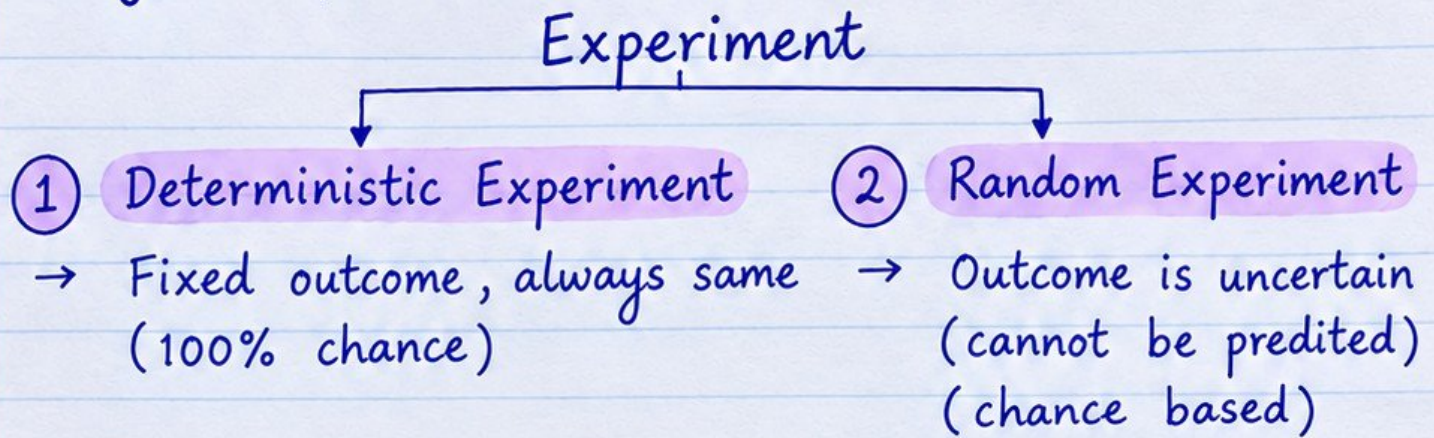


∴ Probability ∴

* Types of Experiments :



* Sample space :

“Set of all possible outcomes of an experiment”.

- Ex :
- Tossing a coin → {H, T}
 - Rolling a die → {1, 2, 3, 4, 5, 6}
 - Tossing a coin → {HH, HT, TH, TT}

* Classical formula :

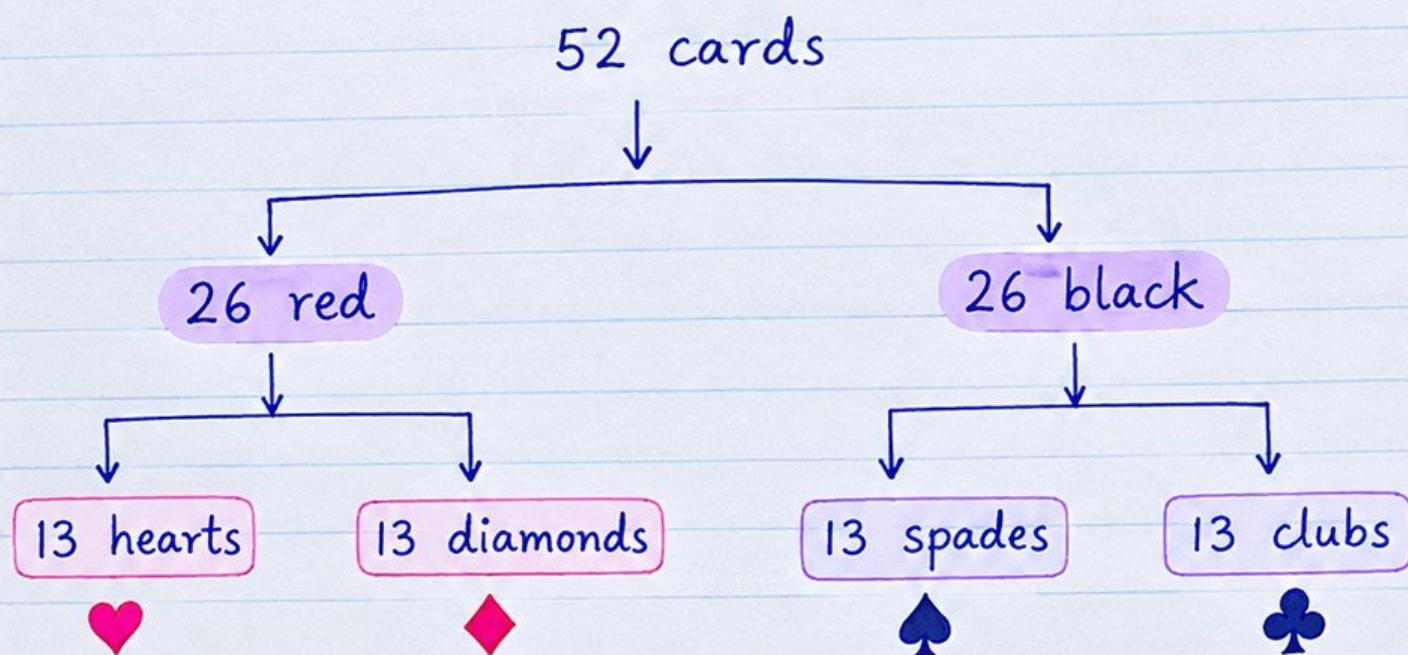
For coin :- $P(\text{Head}) = \frac{1}{2}$ (n times) → $\frac{n}{2}$

For die :- $P(\text{rolling}) = \frac{1}{6}$ (n times) → $\frac{n}{6}$

* Q : How many ways can a die be rolled thrice ?

$$6^3 = 6 \times 6 \times 6 = 36 \times 6 = \boxed{216} \leq$$

Special Case of Cards:



A) $P(\spadesuit \text{ or } \clubsuit)$:

Favourable outcomes = spades or clubs
= $13 + 13 = 26$

Total outcomes = 52

$$P(\spadesuit \text{ or } \clubsuit) = \frac{26}{52} = \boxed{\frac{1}{2}}$$

B) $P(2 - 10)$:

Favourable outcomes = 2, 3, 4, 5, 6, 7, 8, 9, 10
= 9 cards

Total outcomes = 52

$$P(2 - 10) = \boxed{\frac{9}{52}}$$

QUESTION SET

- ① You roll a fair six-sided die. What is the probability of getting an even number?

→ Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Even outcomes: $\{2, 4, 6\}$

$$P(\text{even}) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}} = \frac{3}{6} = \frac{1}{2}$$

- ② 2 coins are tossed. Find the sample space associated with this random experiment.

→ Sample space: $S = \{(H, H), (H, T), (T, H), (T, T)\}$

Total number of cases: $2^n = 2^2 = 4$

- ③ Two six-sided dice are rolled. What is the probability that the sum is 7?

Total number of cases: $6^2 = 36$

Favourable outcomes (sum = 7):

$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

Number of favourable outcomes = 6

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$



★ Remember: Probability = $\frac{\text{Favourable outcomes}}{\text{Total outcomes}}$



QUESTION SET

- ④ A bag contains 3 red, 2 blue and 5 green balls. Find $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$.

→ Total balls = $3 + 2 + 5 = 10$

$$P(\text{red}) = \frac{\text{No. of red balls}}{\text{Total balls}} = \frac{3}{10} = \frac{3}{10}$$

$$P(\text{blue}) = \frac{\text{No. of blue balls}}{\text{Total balls}} = \frac{2}{10} = \frac{1}{5}$$

$$P(\text{green}) = \frac{\text{No. of green balls}}{\text{Total balls}} = \frac{5}{10} = \frac{1}{2}$$

- ⑤ A card is drawn at random from a deck of 52 cards. Find the probability of:

① getting a heart

② getting a face card

→ Total cards = 52

$$\textcircled{1} P(\text{heart}) = \frac{\text{No. of heart cards}}{\text{Total cards}} = \frac{13}{52} = \frac{1}{4}$$

• There are 13 heart cards in a deck.

$$\textcircled{2} P(\text{face card}) = \frac{\text{No. of face cards}}{\text{Total cards}} = \frac{4 \times 3}{52} = \frac{12}{52} = \frac{3}{13}$$

• Face cards are J, Q, K in each suit.

• So, 3 face cards $\times$ 4 suits = 12 face cards.

$$\frac{12}{52} = \frac{3}{13}$$



Remember : Probability = $\frac{\text{Favourable outcomes}}{\text{Total outcomes}}$

